VAANYA KSHATRIYA

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Portfolio: **vaanyas-portfolio-meah.vercel.app**

SUMMARY

Third-year CS student at MIT-WPU (AI & Data Science, CGPA 9.22), looking for internships in AI, ML and Data Science. I build pipelines, train models, and deploy them across fraud detection, medical imaging and computer vision. Have a fraud detection research manuscript under review at the Journal of Young Investigators. VP, Synapse AI Club.

EDUCATION

Bachelor of Technology in Computer Science and Engineering — **2024 to 2028**
Specialisation: Artificial Intelligence and Data Science | MIT World Peace University
Current CGPA: 9.22 | Semester GPA: 9.5

SKILLS

Languages	Python, C++, SQL, HTML, CSS, JavaScript
Frameworks and Tools	PyTorch, XGBoost, FastAPI, React, Node.js, Pandas, NumPy, OpenCV, Git, GitHub, Power BI
Core Competencies	Machine Learning, Computer Vision, Data Analysis, EDA, Data Visualisation, Full-Stack Development
Skills	Agentic AI Tools: Claude Code, GitHub Copilot, Cursor
Interests	AI/ML research, Building Data Products, Entrepreneurship and Leadership Roles

PROJECTS

Credit Card Fraud Detection — **2025 to Present**
Python, scikit-learn, XGBoost, Pandas, SMOTE, Feature Engineering | *Fintech* |
github.com/vaanyas-codec/credit-card-fraud-detection

- Achieved 91.23% F1 and 99.31% ROC-AUC (Receiver Operating Characteristic - Area Under Curve) on a 0.17% fraud-rate dataset by building a leakage-free pipeline with SMOTE (Synthetic Minority Oversampling Technique) oversampling, feature scaling, and outlier capping across 4 classifiers validated by 5-fold cross-validation.
- Built an interactive dashboard to visualise model performance and decision-threshold tradeoffs. Research manuscript under review at the Journal of Young Investigators.

Skin Lesion Classifier — **2026**
PyTorch, EfficientNet-B0, FastAPI, React, Grad-CAM | *MedTech* |
skin-lesion-classifier-seven.vercel.app

- Improved melanoma recall from 60% to 72% on a 58:1 imbalanced dataset by replacing accuracy-optimised training with focal loss on EfficientNet-B0 fine-tuned across 10K HAM10000 images.
- Deployed a full-stack skin lesion classifier (FastAPI + React) using Grad-CAM saliency maps to confirm model attention on lesion regions, not background artefacts.

Real-Time ASL Detection **2026**

Python, OpenCV, MediaPipe, scikit-learn, Claude API | *AccessibilityTech* |
https://github.com/vaanyas-codec/realtime-sign-language-recognition

- Built a 94.5%-accurate real-time ASL (American Sign Language) recogniser across 28 words using MediaPipe + sklearn, iterating through 3 architectures to solve a live-accuracy collapse at scale.
- Tracked down a live-accuracy collapse when the vocabulary scaled up by testing several hypotheses one by one, ruling out mirroring and feature-scale bugs before landing on the real cause: not enough variety in the training clips.
- Built a "collect mode" to record training data through the same live pipeline used for recognition, reaching 94.5% cross-validated accuracy across all 28 words.
- Integrated Claude API to translate recognised ASL signs into natural language; replaced with a lightweight local model due to API cost constraints.

EXPERIENCE

Intern at Accodyn Technologies **Feb 2025 to June 2025**

Frontend Development and Project Management

- Started on frontend, designing and building interfaces for Credit Kit and Credveda across 10+ screens for 4-5 NBFC clients.
- Moved into project management, coordinating a team of 6-8 across frontend, backend, database, and API workstreams; tracking deliverables, planning sprints, and translating client requirements into tasks.

EXTRACURRICULARS

Vice President, Synapse AI Club, MIT-WPU **2024 to Present**

Tech Team Member (2024), Treasurer (2025), Vice President (2026 onwards)

- Joined as a Technical Team Member in 2024, contributing to projects including DriveOpt. Became Treasurer in 2025, managing club finances across workshops, hackathons, and major events. Now serving as Vice President, leading club initiatives and technical sessions for a community of 500+ members.

Thrive Head, E-Cell, MIT-WPU **2024 to Present**

Member (2025), HOD Collaboration at MahaSummit, Core Organiser at RIDE, Thrive Head (2026 onwards)

- Joined E-Cell as a member in 2025, led Collaboration and Convention at MahaSummit, then joined the core organising body of RIDE as one of the main organisers. Managed end-to-end execution, mentor coordination, and participant milestones for the university's largest entrepreneurship event. This work contributed to MIT-WPU's recognition in the World Book of Records and the Asia Book of Records.

Member, Institution's Innovation Council (IIC) **2026 to Present**

Secretary, RIDE Division 1

- Contributing to campus innovation and entrepreneurship initiatives through the IIC; serving as Secretary for RIDE Division 1.